MIL-E-1/652B
1 April 1968
SUPERSEDING
MIL-E-1/652A
21 January 1959

MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 9001

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, UHF

| | | |
|---|---|---|
| Outline | --- | 5-1 (EIA) |
| Base | --- | E7-1 |
| Envelope | --- | T5-1/2 |
| Cathode | --- | Coated unipotential |

Base connections:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Element | --- | g1 | k | h | h | p | g2 | k, g3, sd |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec2 | Ec1 | Pg2 | Pp | Ehk | Alt |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | W | W | v | ft |
| Maximum: | 6.9 (see note 1) | 275 | 110 | --- | 0.11 | 0.55 | 90 | (see note 2) |
| Minimum: | 5.7 | --- | --- | -3.0 | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 | 250 | 100 | -3.0 | --- | --- | --- | --- |

REQUIREMENT OR TEST:

GENERAL

| | | | |
|---|---|---|---|
| 3.2 | Qualification - Required | E-50.3 | Preheating |
| E-50.2 | Holding period | 5. | Preparation for delivery |

(B) denotes changes



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| METHOD OR PARAGRAPH | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| 1266 | Total grid current | See note 3 | 0.65 | II | Ic1 | 0 | -1.0 | µAdc |
| 1256 | Electrode current (1) (anode) | | 0.65 | II | Ib | 1.0 | 3.0 | mAdc |
| 1256 | Electrode current (2) (anode) | Ec1 = -10 Vdc | 0.65 | II | Ib | 0 | 25 | µAdc |
| 1306 | Transconductance (1) | | 0.65 | II | Sm | 1,000 | 1,800 | µmhos |
| 1231 | Emission current | Eb = Ec1 = Ec2 = 15 Vdc (see note 3) | 0.65 | II | Is | 20 | --- | mAdc |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| Appendix D, 20(a), 30 | Visual and mechanical inspection criteria | | --- | --- | --- | --- | --- | --- |
| Appendix D, 20(b) | Dimensions | Outline 5-1; dimensions A thru E | 1.0 | I | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1031 | Low-frequency vibration | Rp = 10,000 ohms | 6.5 | L6 | Ep | --- | 300 | mVac |
| (B) 1301 | Heater current | | 6.5 | L6 | If | 140 | 160 | mA |
| (B) 1336 | Heater-cathode leakage | | 6.5 | L6 | Ihk | --- | 20 | µAdc |
| 1256 | Screen-grid current | | 6.5 | L6 | Ic2 | 0.3 | 1.25 | mAdc |
| 1306 | Transconductance (2) | Ef = 5.5 V | 6.5 | L6 | Sm | 850 | --- | µmhos |
| 1211 | Insulation of electrodes | | 4.0 | L6 | R | 10 | --- | Meg |
| 1331 | Direct-interelectrode capacitance | Shield No. 316<br>Without shield<br>Without shield | 6.5 | L6 | Cgp<br>Cin<br>Cout | ---<br>2.9<br>2.1 | 0.010<br>4.3<br>3.9 | pf<br>pf<br>pf |
| 1121 and 4.3.2 | Base strain | | --- | --- | --- | --- | --- | --- |
| 2126 | Glass strain | | 4.0 | I | --- | --- | --- | --- |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| 1501 and 4.7 | Life-test provisions (intermittent) | Group A;<br>Ehk = 90 Vdc | --- | --- | --- | --- | --- | --- |
| 4.7.3 | Life-test end point (intermittent) (500 hours) | Transconductance (1) | --- | --- | Sm | 800 | --- | µmhos |



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NOTES:

1. Operation of type 9001 in series with other types on supply voltages exceeding 32 volts is not recommended due to the excessive filament surge voltages which appear on tubes of this type.
2. See 6.5.2.3 for reduced pressure (altitude) rating, and for altitude, maximum peak voltage. (B)
3. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Navy - EC

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